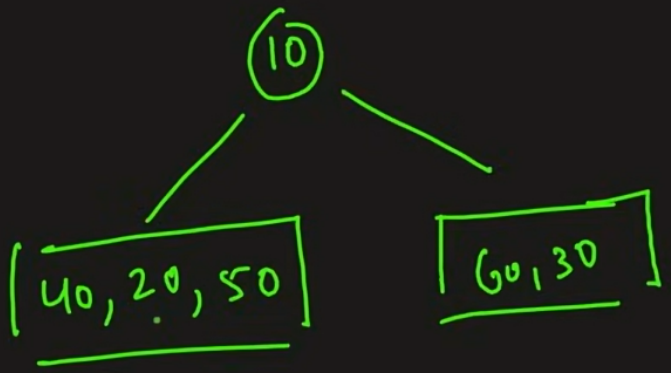
2.40

Construct Binary Tree from Inorder & Preorder



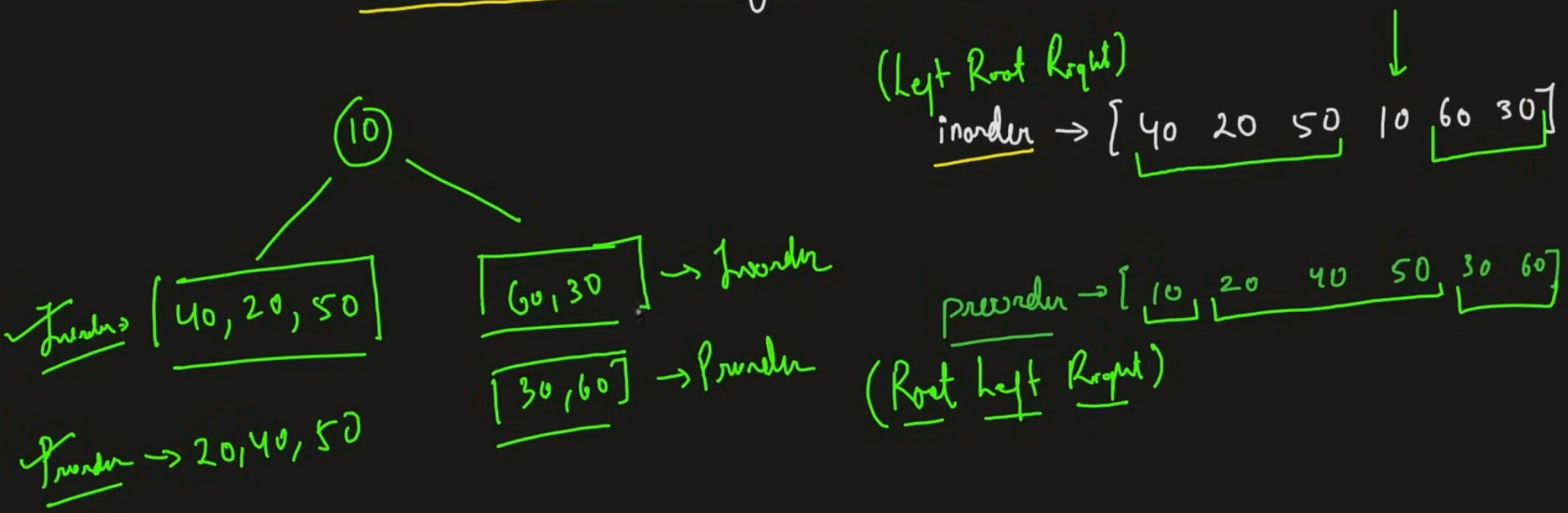
(Left Root Right)

inorder → [40 20 50 10 60 30]

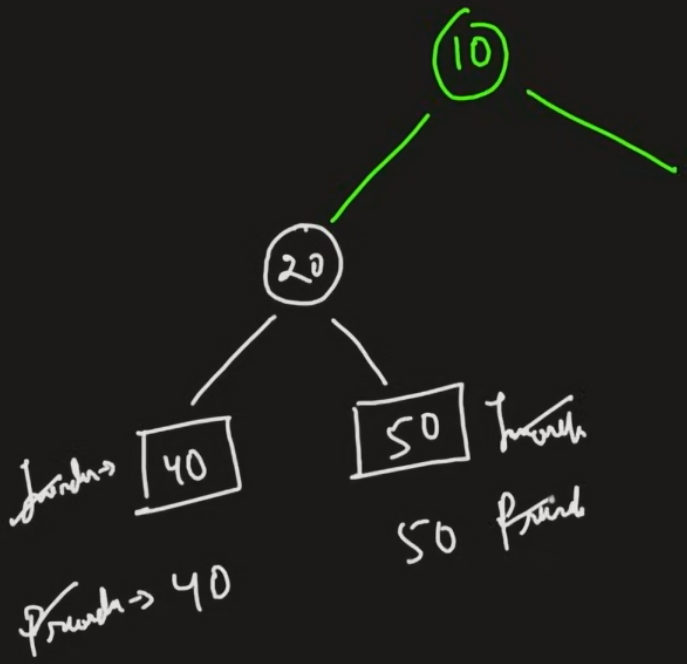
preorder → [10 20 40 50 30 60]

(Root Left Right)

Construct Binary Tree from Inorder & Preorder



Construct Binary Tree from Inorder & Preorder



Left Root Right

Inorder → [40, 20, 50]

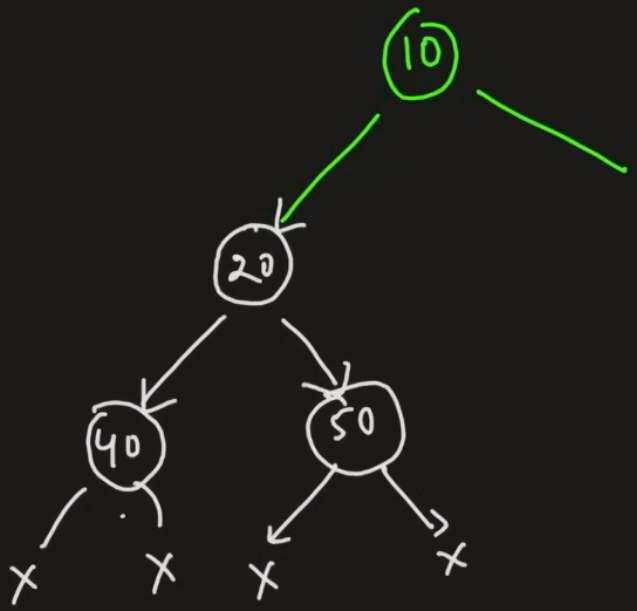
Preorder → 20, 40, 50

Root Left Right

{ [60, 30] → Inorder

[30, 60] → Preorder

Construct Binary Tree from Inorder & Preorder



→ Inorder → 40

Preorder → 40

50 Inorder

50 Preorder

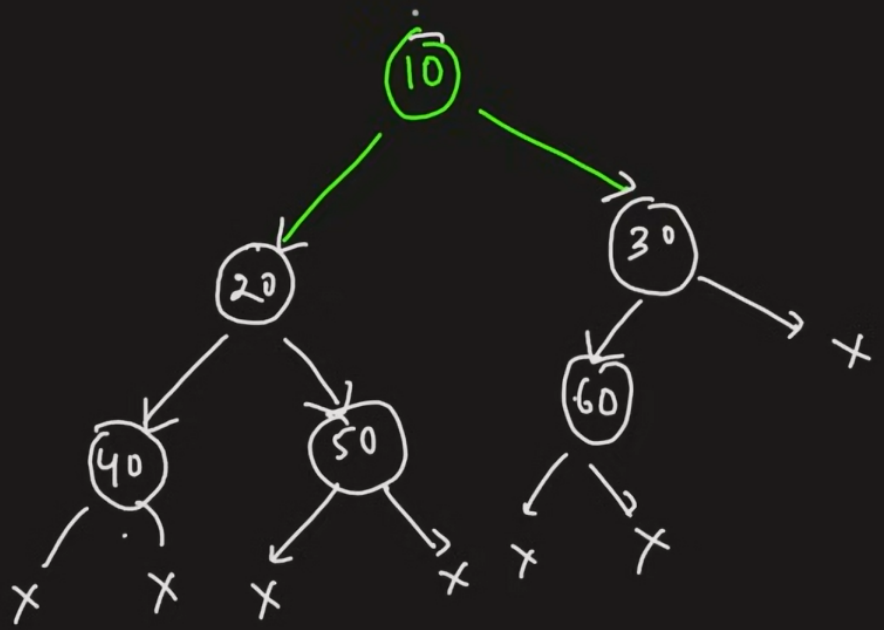
→

60, 30 → Inorder

30, 60 → Preorder

2.40

Construct Binary Tree from Inorder & Preorder



→

{

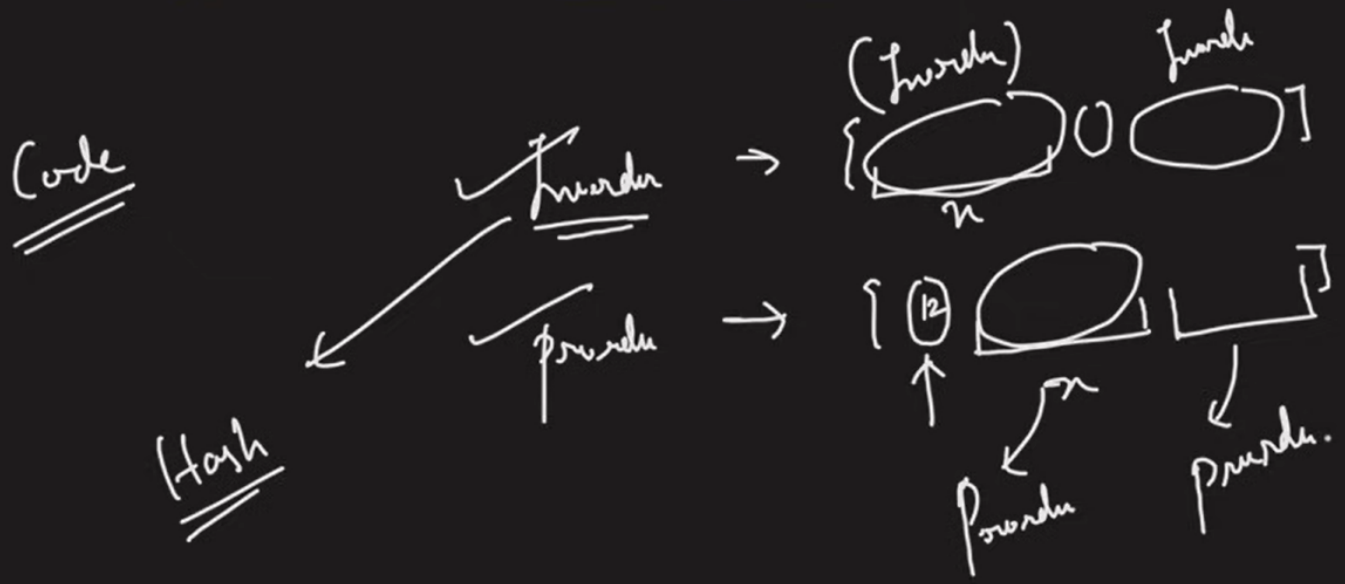
60, 30 → Inorder

30, 60 → Preorder

Inorder → 60

Preorder → 60

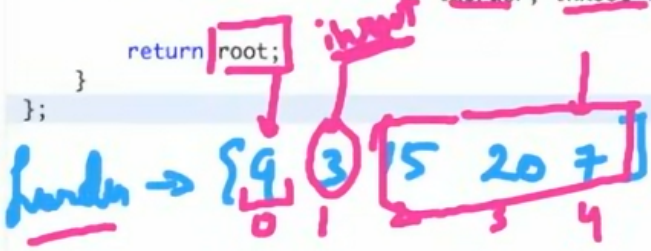
Construct Binary Tree from Inorder & Preorder

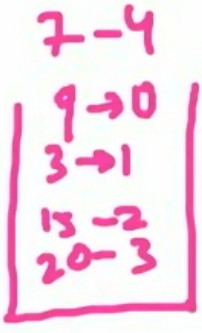


L34. Construct a Binary Tree from Preorder and Inorder Traversal | C++ | Java

C++ Autocomplete

```cpp
class Solution {
public:
    TreeNode* buildTree(vector<int>& preorder, vector<int>& inorder) {
        map<int, int> inMap;

        for(int i = 0; i < inorder.size(); i++) {
            inMap[inorder[i]] = i;
        }

        TreeNode* root = buildTree(preorder, 0, preorder.size() - 1,
                                   inorder, 0, inorder.size() - 1, inMap);
        return root;
    }
    TreeNode* buildTree(vector<int>& preorder, int preStart, int preEnd,
                        vector<int>& inorder,
                        int inStart, int inEnd, map<int, int> inMap) {
        if(preStart > preEnd || inStart > inEnd) return NULL;

        TreeNode* root = new TreeNode(preorder[preStart]);

        int inRoot = inMap[root->val];
        int numsLeft = inRoot - inStart;

        root->left = buildTree(preorder, preStart + 1, preStart + numsLeft,
                               inorder, inStart, inRoot - 1, inMap);
        root->right = buildTree(preorder, preStart + numsLeft + 1, preEnd,
                                inorder, inRoot + 1, inEnd, inMap);

        return root;
    }
};
```





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